

EXERCISE 2.4

Q.1 Find $\frac{dy}{dx}$ by making suitable substitutions in the following functions defined as:

(i) $y = \sqrt{\frac{1-x}{1+x}}$

Solution:

Let $t = \frac{1-x}{1+x}$

then $y = \sqrt{t}$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}} \dots (i)$$

$$\frac{dy}{dt} = \frac{1}{2} \sqrt{\frac{1+x}{1-x}}$$

Now $\frac{dt}{dx} = \frac{(1+x)(-1) - (1-x)(1)}{(1+x)^2}$

$$\frac{dt}{dx} = \frac{-1 - \cancel{1} - 1 + \cancel{1}}{(1+x)^2}$$

$$\frac{dt}{dx} = \frac{-2}{(1+x)^2} \dots (ii)$$

Applying chain rule on (i) & (ii)

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{1}{2} \sqrt{\frac{1+x}{1-x}} \cdot \frac{-2}{(1+x)^2}$$

$$\boxed{\frac{dy}{dx} = -\frac{1}{\sqrt{1-x} \cdot (1+x)^{\frac{3}{2}}}}$$

(ii) $y = \sqrt{x+\sqrt{x}}$

Solution:

Let $t = x + \sqrt{x}$

then $y = \sqrt{t}$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}}$$

$$\frac{dy}{dt} = \frac{1}{2} \frac{1}{\sqrt{x+\sqrt{x}}} \dots (i)$$

also, $t = x + \sqrt{x}$

$$\frac{dt}{dx} = 1 + \frac{1}{2\sqrt{x}} \dots (ii)$$

Applying chain rule on equation (i) and equation (ii),

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x+\sqrt{x}}} \cdot \frac{1+2\sqrt{x}}{2\sqrt{x}}$$

$$\boxed{\frac{dy}{dx} = \frac{1+2\sqrt{x}}{4\sqrt{x}\sqrt{x+\sqrt{x}}}}$$

(iii) $y = x\sqrt{\frac{a+x}{a-x}}$

Solution:

Let $t = \frac{a+x}{a-x}$ then $y = xt^{\frac{1}{2}}$

$$t = \frac{a+x}{a-x}$$

$$\frac{dt}{dx} = \frac{(a-x)(1) - (a+x)(-1)}{(a-x)^2}$$

$$\frac{dt}{dx} = \frac{a - \cancel{x} + a + \cancel{x}}{(a-x)^2}$$

$$\frac{dt}{dx} = \frac{2a}{(a-x)^2} \dots (i)$$

Now, $y = xt^{\frac{1}{2}}$

$$\frac{dy}{dt} = x \cdot \frac{1}{2\sqrt{t}} + \sqrt{t} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{x}{2} \cdot \frac{1}{\sqrt{t}} + \sqrt{t} \cdot \frac{(a-x)^2}{2a}$$

$$\frac{dy}{dt} = \frac{x}{2} \cdot \frac{\sqrt{a-x}}{\sqrt{a+x}} + \sqrt{\frac{a+x}{a-x}} \cdot \frac{(a-x)^2}{2a} \dots (ii)$$

Applying chain rule on equation (i) and equation (ii).

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dx} = \left[\frac{x\sqrt{a-x}}{2\sqrt{a+x}} + \frac{\sqrt{a+x}(a-x)^2}{\sqrt{a-x} \cdot 2a} \right] \frac{2a}{(a-x)^2}$$

$$\frac{dy}{dx} = \left[\frac{ax(a-x) + (a+x)(a-x)^2}{2a\sqrt{a+x}\sqrt{a-x}} \right] \frac{2a}{(a-x)^2}$$

$$\frac{dy}{dx} = (a-x) \left[\frac{ax + a^2 - x^2}{\sqrt{a+x}\sqrt{a-x}} \right] \frac{1}{(a-x)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{a^2 - x^2 + ax}{(a+x)^{\frac{1}{2}}(a-x)^{\frac{3}{2}}}}$$

(iv) $y = (3x^2 - 2x + 7)^6$

Solution:

Let $t = 3x^2 - 2x + 7$

$$\frac{dt}{dx} = 6x - 2 \dots (i)$$

Also, $y = t^6 \Rightarrow \frac{dy}{dt} = 6t^5$

$$\Rightarrow \frac{dy}{dt} = 6(3x^2 - 2x + 7)^5 \dots (ii)$$

Applying chain rule on equation (i) and equation (ii)

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= 6(3x^2 - 2x + 7)^5 \cdot (6x - 2)$$

$$\boxed{\frac{dy}{dx} = 6(6x - 2)(3x^2 - 2x + 7)^5}$$

(v) $y = \sqrt{\frac{a^2 + x^2}{a^2 - x^2}}$

Solution:

Let $t = \frac{a^2 + x^2}{a^2 - x^2}$

$$\Rightarrow y = \sqrt{t}$$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}}$$

$$\frac{dy}{dt} = \frac{1}{2} \sqrt{\frac{a^2 - x^2}{a^2 + x^2}} \dots (i)$$

$$t = \frac{a^2 + x^2}{a^2 - x^2}$$

$$\frac{dt}{dx} = \frac{(a^2 - x^2)(2x) - (a^2 + x^2)(-2x)}{(a^2 - x^2)^2}$$

$$\frac{dt}{dx} = \frac{\cancel{2a^2}x - 2x^3 + \cancel{2a^2}x + 2x^3}{(a^2 - x^2)^2}$$

$$\frac{dt}{dx} = \frac{4a^2x}{(a^2 - x^2)^2} \dots (ii)$$

Applying chain rule on equation (i) and equation (ii)

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{1}{2} \sqrt{\frac{a^2 - x^2}{a^2 + x^2}} \cdot \frac{4a^2x}{(a^2 - x^2)^2}$$

$$\frac{dy}{dx} = \frac{2a^2x}{\sqrt{a^2 + x^2}(a^2 - x^2)^{\frac{3}{2}}}$$

Q.2 Find $\frac{dy}{dx}$ if:

(i) $3x + 4y + 7 = 0$

Solution:

$$3x + 4y + 7 = 0$$

Differentiate w.r.t "x"

$$3 + 4 \frac{dy}{dx} = 0$$

$$4 \frac{dy}{dx} = -3$$

$$\boxed{\frac{dy}{dx} = \frac{-3}{4}}$$

(ii) $xy + y^2 = 2$

Solution:

$$xy + y^2 = 2$$

Differentiate w.r.t "x"

$$x \frac{dy}{dx} + y(1) + 2y \frac{dy}{dx} = 0$$

$$(x + 2y) \frac{dy}{dx} = -y$$

$$\boxed{\frac{dy}{dx} = \frac{-y}{x + 2y}}$$

(iii) $x^2 - 4xy - 5y = 0$

Solution:

$$x^2 - 4xy - 5y = 0$$

Differentiate w.r.t "x"

$$2x - 4 \left[x \frac{dy}{dx} + y(1) \right] - 5 \frac{dy}{dx} = 0$$

$$2x - 4x \frac{dy}{dx} - 4y - 5 \frac{dy}{dx} = 0$$

$$(4x+5) \frac{dy}{dx} = 2(x-2y)$$

$$\boxed{\frac{dy}{dx} = \frac{2(x-2y)}{4x+5}}$$

(iv) $4x^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Solution:

$$4x^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

Differentiate w.r.t "x".

$$8x + 2h \left[x \frac{dy}{dx} + y \right] + 2by \frac{dy}{dx} + 2g + 2f \frac{dy}{dx} = 0$$

$$2hx \frac{dy}{dx} + 2by \frac{dy}{dx} + 2f \frac{dy}{dx} = -8x - 2hy - 2g$$

$$\cancel{2} (hx + by + f) \frac{dy}{dx} = -\cancel{2} (4x + hy + g)$$

$$\boxed{\frac{dy}{dx} = -\frac{4x + hy + g}{hx + by + f}}$$

(v) $x\sqrt{1+y} + y\sqrt{1+x} = 0$

Solution:

$$x\sqrt{1+y} + y\sqrt{1+x} = 0$$

Differentiate w.r.t "x",

$$= \frac{d}{dx} (x\sqrt{1+y} + y\sqrt{1+x}) = \frac{d}{dx} (0)$$

$\Rightarrow$

$$x \frac{d}{dx} \sqrt{1+y} + \sqrt{1+y} \frac{d}{dx} (x) + y \frac{d}{dx} \sqrt{1+x} + \sqrt{1+x} \frac{dy}{dx} = 0$$

$$x \frac{1}{2\sqrt{1+y}} \cdot \frac{dy}{dx} + \sqrt{1+y} (1) + y \frac{1}{2\sqrt{1+x}} + \sqrt{1+x} \frac{dy}{dx} = 0$$

$$\left(\frac{x}{2\sqrt{1+y}} + \sqrt{1+x} \right) \frac{dy}{dx} = - \left[\sqrt{1+y} + \frac{y}{2\sqrt{1+x}} \right]$$

$$\left(\frac{x+2\sqrt{(1+x)(1+y)}}{\cancel{2}\sqrt{1+y}} \right) \frac{dy}{dx} = - \left[\frac{2\sqrt{(1+x)(1+y)} + y}{\cancel{2}\sqrt{1+x}} \right]$$

$$\frac{x+2\sqrt{1+x+y+xy}}{\sqrt{1+y}} \frac{dy}{dx} = - \frac{y+2\sqrt{1+x+y+xy}}{\sqrt{1+x}}$$

$$\boxed{\frac{dy}{dx} = - \frac{(y+2\sqrt{1+x+y+xy})(\sqrt{1+y})}{(x+2\sqrt{1+x+y+xy})(\sqrt{1+x})}}$$

(vi) $y(x^2-1) = x\sqrt{x^2+4}$

Solution:

$$y(x^2-1) = x\sqrt{x^2+4}$$

Differentiate w.r.t "x",

$$\frac{d}{dx} (y(x^2-1)) = \frac{d}{dx} (x(\sqrt{x^2+4}))$$

$$y \frac{d}{dx} (x^2-1) + (x^2-1) \frac{d}{dx} (y) = x \frac{d}{dx} \sqrt{x^2+4} + \sqrt{x^2+4} \frac{d}{dx} (x)$$

$$y(2x) + (x^2-1) \frac{dy}{dx} = x \frac{1(2x)}{2\sqrt{x^2+4}} + \sqrt{x^2+4} (1)$$

$$2xy + (x^2-1) \frac{dy}{dx} = \frac{x^2}{\sqrt{x^2+4}} + \sqrt{x^2+4}$$

$$(x^2-1) \frac{dy}{dx} = \frac{x^2 + x^2 + 4}{\sqrt{x^2+4}} - 2xy$$

$$(x^2-1) \frac{dy}{dx} = \frac{2x^2 + 4 - 2xy(\sqrt{x^2+4})}{\sqrt{x^2+4}}$$

$$(x^2-1) \frac{dy}{dx} = \frac{2(x^2+2) - 2xy\sqrt{x^2+4}}{\sqrt{x^2+4}}$$

$$\boxed{\frac{dy}{dx} = \frac{2(x^2+2) - 2xy\sqrt{x^2+4}}{\sqrt{x^2+4}(x^2-1)}}$$

Q.3 Find $\frac{dy}{dx}$ for the following

parametric functions:

(i) $x = \theta + \frac{1}{\theta}$ and $y = \theta + 1$

Solution:

$$x = \theta + \frac{1}{\theta}$$

Differentiate w.r.t "θ",

$$\frac{dx}{d\theta} = 1 - \frac{1}{\theta^2} \Rightarrow \frac{dx}{d\theta} = \frac{\theta^2 - 1}{\theta^2} \dots (i)$$

$$y = \theta + 1$$

Differentiate w.r.t “ θ ”,

$$\frac{dy}{d\theta} = 1 \dots (ii)$$

Applying chain rule on equation (i) and equation (ii)

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$\frac{dy}{dx} = 1 \cdot \frac{\theta^2}{\theta^2 - 1}$$

$$\boxed{\frac{dy}{dx} = \frac{\theta^2}{\theta^2 - 1}}$$

(ii) $x = \frac{a(1-t^2)}{1+t^2}, y = \frac{2bt}{1+t^2}$

Solution:

$$x = \frac{a(1-t^2)}{1+t^2}$$

Differentiate w.r.t “ t ”,

$$\frac{dx}{dt} = a \frac{(1+t^2)(-2t) - (1-t^2)(2t)}{(1+t^2)^2}$$

$$= a \left[\frac{-2t - \cancel{2t^3} - 2t + \cancel{2t^3}}{(1+t^2)^2} \right]$$

$$\frac{dx}{dt} = \frac{-4at}{(1+t^2)^2} \dots (i)$$

$$y = \frac{2bt}{1+t^2}$$

Differentiate w.r.t “ t ”

$$\frac{dy}{dt} = \frac{(1+t^2)(2b) - 2bt(2t)}{(1+t^2)^2}$$

$$= \frac{2b + 2bt^2 - 4bt^2}{(1+t^2)^2}$$

$$= \frac{2b - 2bt^2}{(1+t^2)^2}$$

$$\frac{dy}{dt} = \frac{2b(1-t^2)}{(1+t^2)^2} \dots (ii)$$

Applying chain rule on equation (i) and equation (ii)

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{-2b(1-t^2)}{(1+t^2)^2} \cdot \frac{(1+t^2)^2}{4at}$$

$$\boxed{\frac{dy}{dx} = \frac{-b(1-t^2)}{2at}}$$

Q.4 Prove that $y \frac{dy}{dx} + x = 0$ if

$$x = \frac{1-t^2}{1+t^2}, y = \frac{2t}{1+t^2}$$

Solution:

$$x = \frac{1-t^2}{1+t^2}$$

Differentiate w.r.t “ t ”,

$$\frac{dx}{dt} = \frac{(1+t^2)(-2t) - (1-t^2)(2t)}{(1+t^2)^2}$$

$$= \frac{-2t - \cancel{2t^3} - 2t + \cancel{2t^3}}{(1+t^2)^2}$$

$$\frac{dx}{dt} = \frac{-4t}{(1+t^2)^2} \dots (i)$$

$$y = \frac{2t}{1+t^2}$$

Differentiate w.r.t “ t ”,

$$\frac{dy}{dt} = \frac{(1+t^2)(2) - (2t)(2t)}{(1+t^2)^2}$$

$$= \frac{2 + 2t^2 - 4t^2}{(1+t^2)^2}$$

$$\frac{dy}{dt} = \frac{2(1-t^2)}{(1+t^2)^2} \dots (ii)$$

Applying chain rule on equation (i) and equation (ii)

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{-2(1-t^2)}{(1+t^2)^2} \cdot \frac{(1+t^2)^2}{4t}$$

$$\frac{dy}{dx} = \frac{-(1-t^2)}{2t}$$

$$\Rightarrow y \frac{dy}{dx} = -\frac{1-t^2}{2t} \times \frac{2t}{1+t^2}$$

$$y \frac{dy}{dx} = -\frac{1-t^2}{1+t^2}$$

$$y \frac{dy}{dx} = -x \quad \because x = \frac{1-t^2}{1+t^2}$$

$$y \frac{dy}{dx} + x = 0$$

Hence the proof.

Q.5 Differentiate

(i) $x^2 - \frac{1}{x^2}$ w.r.t. x^4

Solution:

$$\text{Let } y = x^2 - \frac{1}{x^2}, \quad t = x^4$$

$$\text{We have to find } \frac{dy}{dt}$$

$$y = x^2 - \frac{1}{x^2}$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = 2x + \frac{2}{x^3}$$

$$\frac{dy}{dx} = \frac{2x^4 + 2}{x^3} \dots (i)$$

$$\text{Also } t = x^4$$

Differentiate w.r.t “x”

$$\frac{dt}{dx} = 4x^3 \dots (ii)$$

Applying chain rule on equation (i) and equation (ii),

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{2(x^4 + 1)}{x^3} \cdot \frac{1}{4x^3}$$

$$\frac{dy}{dt} = \frac{2(x^4 + 1)}{4x^6}$$

$$\boxed{\frac{dy}{dt} = \frac{x^4 + 1}{2x^6}}$$

(ii) $(1+x^2)^n$ w.r.t x^2

Solution:

$$\text{Let } y = (1+x^2)^n, \quad t = x^2$$

$$\text{We have to find } \frac{dy}{dt}$$

$$y = (1+x^2)^n$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = n(1+x^2)^{n-1} (2x)$$

$$\frac{dy}{dx} = 2nx(1+x^2)^{n-1} \dots (i)$$

$$\text{also } t = x^2$$

Differentiate w.r.t “x”

$$\frac{dt}{dx} = 2x \dots (ii)$$

Applying chain rule on equation (i) and equation (ii)

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = \cancel{2n} (1+x^2)^{n-1} \cdot \cancel{2} \cdot \frac{1}{\cancel{2}}$$

$$\boxed{\frac{dy}{dt} = n(1+x^2)^{n-1}}$$

(iii) $\frac{x^2+1}{x^2-1}$ w.r.t $\frac{x-1}{x+1}$

Solution:

$$\text{Let } y = \frac{x^2+1}{x^2-1}, \quad t = \frac{x-1}{x+1}$$

We have to find $\frac{dy}{dt}$

$$y = \frac{x^2 + 1}{x^2 - 1}$$

Differentiate w.r.t "x"

$$\frac{dy}{dx} = \frac{(x^2 - 1)(2x) - (x^2 + 1)(2x)}{(x^2 - 1)^2}$$

$$\frac{dy}{dx} = \frac{\cancel{2x^3} - 2x - \cancel{2x^3} - 2x}{(x^2 - 1)^2}$$

$$\frac{dy}{dx} = \frac{-4x}{(x^2 - 1)^2} \dots (i)$$

$$\text{also } t = \frac{x-1}{x+1}$$

Differentiate w.r.t "x"

$$\frac{dt}{dx} = \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2}$$

$$= \frac{\cancel{x} + 1 - \cancel{x} + 1}{(x+1)^2}$$

$$\frac{dt}{dx} = \frac{2}{(x+1)^2} \dots (ii)$$

Applying chain rule on equation (i)

and equation (ii)

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{-4x}{(x^2 - 1)^2} \cdot \frac{(x+1)^2}{2}$$

$$\frac{dy}{dt} = -\frac{2x(x+1)^2}{(x+1)^2(x-1)^2}$$

$$\boxed{\frac{dy}{dt} = \frac{-2x}{(x-1)^2}}$$

(iv) $\frac{ax+b}{cx+d}$ w.r.t $\frac{ax^2+b}{ax^2+d}$

Solution:

$$\text{Let } y = \frac{ax+b}{cx+d}, \quad t = \frac{ax^2+b}{ax^2+d}$$

we have to find $\frac{dy}{dt}$

$$y = \frac{ax+b}{cx+d}$$

Differentiate w.r.t "x"

$$\frac{dy}{dx} = \frac{(cx+d)(a) - (ax+b)(c)}{(cx+d)^2}$$

$$= \frac{\cancel{acx} + ad - \cancel{acx} - bc}{(cx+d)^2}$$

$$\frac{dy}{dx} = \frac{ad-bc}{(cx+d)^2} \dots (i)$$

$$\text{also } t = \frac{ax^2+b}{ax^2+d}$$

Differentiate w.r.t "x"

$$\frac{dt}{dx} = \frac{(ax^2+d)(2ax) - (ax^2+b)(2ax)}{(ax^2+d)^2}$$

$$\frac{dt}{dx} = \frac{\cancel{2a^2x^3} + 2adx - \cancel{2a^2x^3} - 2abx}{(ax^2+d)^2}$$

$$\frac{dt}{dx} = \frac{2ax(d-b)}{(ax^2+d)^2} \dots (ii)$$

Applying chain rule on equation (i)

and equation (ii)

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{ad-bc}{(cx+d)^2} \cdot \frac{(ax^2+d)^2}{2ax(d-b)}$$

$$\boxed{\frac{dy}{dt} = \frac{(ad-bc)(ax^2+d)^2}{2ax(d-b)(cx+d)^2}}$$

(v) $\frac{x^2+1}{x^2-1}$ w.r.t x^3

Solution:

Let $y = \frac{x^2+1}{x^2-1}, \quad t = x^3$

we have to find $\frac{dy}{dt}$

$$y = \frac{x^2+1}{x^2-1}$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = \frac{(x^2-1)(2x) - (x^2+1)(2x)}{(x^2-1)^2}$$

$$\frac{dy}{dx} = \frac{\cancel{2x^3} - 2x - \cancel{2x^3} - 2x}{(x^2-1)^2}$$

$$\frac{dy}{dx} = \frac{-4x}{(x^2-1)^2} \dots (i)$$

also $t = x^3$

Differentiate w.r.t “x”

$$\frac{dt}{dx} = 3x^2 \dots (ii)$$

Applying chain rule on equation (i) and equation (ii)

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{-4x}{(x^2-1)^2} \cdot \frac{1}{3x^2}$$

$$\frac{dy}{dt} = \frac{-4}{3x(x^2-1)^2}$$

DERIVATIVES OF TRIGONOMETRIC FUNCTIONS:

$$(1) \quad \frac{d}{dx}(\sin x) = \cos x$$

$$(2) \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$(3) \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

$$(4) \quad \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$(5) \quad \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$(6) \quad \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$(1) \quad \text{Prove that: } \frac{d}{dx}(\sin x) = \cos x$$

$$\text{Let } y = \sin x$$

$$y + \delta y = \sin(x + \delta x)$$

$$\delta y = \sin(x + \delta x) - \sin x$$

$$= 2 \cos\left(\frac{x + \delta x + x}{2}\right) \sin\left(\frac{x + \delta x - x}{2}\right)$$

$$\because \sin \alpha - \sin \beta = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

$$= 2 \cos\left(x + \frac{\delta x}{2}\right) \sin\left(\frac{\delta x}{2}\right)$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = 2 \cos\left(x + \frac{\delta x}{2}\right) \frac{\sin\left(\frac{\delta x}{2}\right)}{\delta x}$$

$$= \cos\left(x + \frac{\delta x}{2}\right) \frac{\sin\left(\frac{\delta x}{2}\right)}{\left(\frac{\delta x}{2}\right)}$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \cos\left(x + \frac{\delta x}{2}\right) \lim_{\delta x \rightarrow 0} \left[\frac{\sin\left(\frac{\delta x}{2}\right)}{\left(\frac{\delta x}{2}\right)} \right]$$

$$\because \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\frac{dy}{dx} = \cos(x) \times 1$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$(2) \quad \text{Prove that: } \frac{d}{dx}(\cos x) = -\sin x$$

$$\text{Let } y = \cos x$$

$$y + \delta y = \cos(x + \delta x)$$

$$\delta y = \cos(x + \delta x) - \cos x$$

$$\delta y = -2 \sin\left(\frac{x + \delta x + x}{2}\right) \sin\left(\frac{x + \delta x - x}{2}\right)$$

$$\because \cos \alpha - \cos \beta = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

$$\delta y = -2 \sin\left(x + \frac{\delta x}{2}\right) \sin\left(\frac{\delta x}{2}\right)$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = \frac{-2 \sin\left(x + \frac{\delta x}{2}\right) \sin\left(\frac{\delta x}{2}\right)}{\delta x}$$

$$= -\sin\left(x + \frac{\delta x}{2}\right) \frac{\sin\left(\frac{\delta x}{2}\right)}{\left(\frac{\delta x}{2}\right)}$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = -\lim_{\delta x \rightarrow 0} \sin\left(x + \frac{\delta x}{2}\right) \lim_{\delta x \rightarrow 0} \left[\frac{\sin\left(\frac{\delta x}{2}\right)}{\frac{\delta x}{2}} \right]$$

$$\because \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\frac{dy}{dx} = -\sin x \times 1$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

(3) **Prove that:** $\frac{d}{dx}(\tan x) = \sec^2 x$

Let $y = \tan x$

$$y = \frac{\sin x}{\cos x}$$

$$y + \delta y = \frac{\sin(x + \delta x)}{\cos(x + \delta x)}$$

$$\delta y = \frac{\sin(x + \delta x)}{\cos(x + \delta x)} - \frac{\sin x}{\cos x}$$

$$= \frac{\sin(x + \delta x)\cos x - \sin x \cos(x + \delta x)}{\cos(x + \delta x)\cos x}$$

$$\delta y = \frac{\sin(x + \delta x - x)}{\cos(x + \delta x)\cos x}$$

$$\therefore \sin \alpha \cos \beta - \cos \alpha \sin \beta = \sin(\alpha - \beta)$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = \frac{\sin(\delta x)}{\delta x \cos(x + \delta x)\cos x}$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \left[\lim_{\delta x \rightarrow 0} \frac{\sin \delta x}{\delta x} \right] \left[\lim_{\delta x \rightarrow 0} \frac{1}{\cos(x + \delta x)\cos x} \right]$$

$$\frac{dy}{dx} = 1 \times \frac{1}{\cos x \cdot \cos x}$$

$$\frac{d}{dx}(\tan x) = \frac{1}{\cos^2 x}$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

(4) **Prove that:** $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

Let $y = \cot x$

$$y = \frac{\cos x}{\sin x}$$

$$y + \delta y = \frac{\cos(x + \delta x)}{\sin(x + \delta x)}$$

$$\delta y = \frac{\cos(x + \delta x)}{\sin(x + \delta x)} - y$$

$$\begin{aligned} \delta y &= \frac{\cos(x + \delta x)}{\sin(x + \delta x)} - \frac{\cos x}{\sin x} \\ &= \frac{\cos(x + \delta x)\sin x - \cos x \sin(x + \delta x)}{\sin(x + \delta x)(\sin x)}, \end{aligned}$$

$$\therefore \sin \alpha \cos \beta - \cos \alpha \sin \beta = \sin(\alpha - \beta)$$

$$= \frac{\sin[x - (x + \delta x)]}{\sin(x + \delta x)(\sin x)}$$

$$\delta y = \frac{\sin(-\delta x)}{\sin(x + \delta x)\sin x}$$

$$\therefore \sin(-\theta) = -\sin \theta$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = -\frac{\sin \delta x}{\delta x [\sin(x + \delta x)\sin x]}$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = -\lim_{\delta x \rightarrow 0} \frac{\sin \delta x}{\delta x} \cdot \lim_{\delta x \rightarrow 0} \frac{1}{\sin(x + \delta x)\sin x}$$

$$\frac{dy}{dx} = -1 \times \frac{1}{\sin x \cdot \sin x}$$

$$\frac{d}{dx}(\cot x) = -\frac{1}{\sin^2 x}$$

$$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

(5) **Prove that:** $\frac{d}{dx}(\sec x) = \sec x \tan x$

Let $y = \sec x$

$$y + \delta y = \sec(x + \delta x)$$

$$\delta y = \sec(x + \delta x) - y$$

$$\delta y = \sec(x + \delta x) - \sec x$$

$$\begin{aligned} \delta y &= \frac{1}{\cos(x + \delta x)} - \frac{1}{\cos x} \\ &= \frac{\cos x - \cos(x + \delta x)}{\cos(x + \delta x)\cos x} \end{aligned}$$

$$\therefore \cos \alpha - \cos \beta = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$$

$$\delta y = \frac{-2 \sin\left(\frac{x+x+\delta x}{2}\right) \sin\left(\frac{x-(x+\delta x)}{2}\right)}{\cos(x+\delta x) \cos x}$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = \frac{-2 \sin\left(x + \frac{\delta x}{2}\right) \sin\left(\frac{-\delta x}{2}\right)}{(\delta x) \cos(x+\delta x) \cos x}$$

$$\frac{\delta y}{\delta x} = \frac{\sin\left(x + \frac{\delta x}{2}\right) \sin\left(\frac{\delta x}{2}\right)}{\cos(x+\delta x) \cos x \left(\frac{\delta x}{2}\right)}$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \sin\left(x + \frac{\delta x}{2}\right) \lim_{\delta x \rightarrow 0} \frac{\sin\left(\frac{\delta x}{2}\right)}{\frac{\delta x}{2}} \lim_{\delta x \rightarrow 0} \frac{1}{\cos(x+\delta x) \cos x}$$

$$\frac{dy}{dx} = \sin x \times 1 \times \frac{1}{\cos x \cos x}$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

(6) **Prove that:**

$$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

Let $y = \operatorname{cosec} x$

$$y = \frac{1}{\sin x}$$

$$y + \delta y = \frac{1}{\sin(x+\delta x)}$$

$$\delta y = \frac{1}{\sin(x+\delta x)} - y$$

$$\begin{aligned} \delta y &= \frac{1}{\sin(x+\delta x)} - \frac{1}{\sin x} \\ &= \frac{\sin x - \sin(x+\delta x)}{\sin(x+\delta x) \sin x} \end{aligned}$$

$$\therefore \sin \alpha - \sin \beta = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

$$= \frac{2 \cos\left(\frac{x+x+\delta x}{2}\right) \sin\left(\frac{x-(x+\delta x)}{2}\right)}{\sin(x+\delta x) \sin x}$$

$$\delta y = \frac{2 \cos\left(x + \frac{\delta x}{2}\right) \sin\left(-\frac{\delta x}{2}\right)}{\sin(x+\delta x) \sin x}$$

Dividing both sides by ' δx '

$$\frac{\delta y}{\delta x} = \frac{-2 \cos\left(x + \frac{\delta x}{2}\right) \sin\left(\frac{\delta x}{2}\right)}{\delta x \sin(x+\delta x) \sin x}$$

$$\frac{\delta y}{\delta x} = \frac{-\cos\left(x + \frac{\delta x}{2}\right) \sin\left(\frac{\delta x}{2}\right)}{\sin(x+\delta x) \sin x \left(\frac{\delta x}{2}\right)}$$

Taking limit as $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = -\lim_{\delta x \rightarrow 0} \cos\left(x + \frac{\delta x}{2}\right) \cdot \lim_{\delta x \rightarrow 0} \frac{1}{\sin(x+\delta x) \sin x} \times \lim_{\delta x \rightarrow 0} \left(\frac{\sin \frac{\delta x}{2}}{\frac{\delta x}{2}}\right)$$

$$\frac{dy}{dx} = -\cos x \times \frac{1}{\sin x \cdot \sin x} \times 1$$

$$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

DERIVATIVES OF INVERSE TRIGONOMETRIC FUNCTIONS

$$(1) \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

$$(2) \quad \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

$$(3) \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, \quad -\infty < x < \infty$$

$$(4) \quad \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}, \quad -\infty < x < \infty$$

$$(5) \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad x \in [-1, 1]'$$

$$(6) \quad \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}, \quad x \in [-1, 1]'$$

(1) **Prove that:**

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

Let $y = \sin^{-1} x$

$$\sin y = x \quad (i)$$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\sin y) = \frac{d}{dx}(x)$$

$$\cos y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{\cos^2 y}} \\ &= \frac{1}{\sqrt{1 - \sin^2 y}} \end{aligned}$$

Using (i)

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1 - x^2}}$$

(2) **Prove that:**

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1 - x^2}}, \quad -1 < x < 1$$

$$\text{Let } y = \cos^{-1} x$$

$$\cos y = x \quad (i)$$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\cos y) = \frac{d}{dx}(x)$$

$$-\sin y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{-1}{\sin y}$$

$$\frac{dy}{dx} = \frac{-1}{\sqrt{\sin^2 y}}$$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1 - \cos^2 y}}$$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1 - x^2}} \text{ Using (i)}$$

(3) **Prove that:**

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2}, \quad -\infty < x < \infty$$

$$\text{Let } y = \tan^{-1} x$$

$$\tan y = x \quad (i)$$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\tan y) = \frac{d}{dx}(x)$$

$$\sec^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2 y}$$

Using (i)

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2}$$

(4) **Prove that:**

$$\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1 + x^2}, \quad -\infty < x < \infty$$

$$\text{Let } y = \cot^{-1} x$$

$$\cot y = x \quad (i)$$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\cot y) = \frac{d}{dx}(x)$$

$$-\operatorname{cosec}^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{-1}{\operatorname{cosec}^2 y}$$

$$\frac{dy}{dx} = \frac{-1}{1 + \cot^2 y}$$

Using (i)

$$\frac{dy}{dx} = -\frac{1}{1 + x^2}$$

$$\frac{d}{dx}(\cot^{-1} y) = -\frac{1}{1 + x^2}$$

(5) **Prove that:**

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2 - 1}}, \quad x \in [-1, 1]'$$

$$\text{Let } y = \sec^{-1} x$$

$$\sec y = x \quad (i)$$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\sec y) = \frac{d}{dx}(x)$$

$$\sec y \tan y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec y \tan y}$$

$$\frac{dy}{dx} = \frac{1}{\sec y \sqrt{\tan^2 y}}$$

$$\frac{dy}{dx} = \frac{1}{\sec y \sqrt{\sec^2 y - 1}}$$

Using (i)

$$\frac{dy}{dx} = \frac{1}{x \sqrt{x^2 - 1}}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x \sqrt{x^2 - 1}}$$

(6) **Prove that:**

$$\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x| \sqrt{x^2 - 1}}, \quad x \in [-1, 1]'$$

$$\text{Let } y = \operatorname{cosec}^{-1} x$$

$$\operatorname{cosec} y = x \quad (i)$$

Differentiate

w.r.t

'x'

$$\frac{d}{dx}(\operatorname{cosec} y) = \frac{d}{dx}(x)$$

$$-\operatorname{cosec} y \cot y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{-\operatorname{cosec} y \cot y}$$

$$\frac{dy}{dx} = -\frac{1}{\operatorname{cosec} y \sqrt{\cot^2 y}}$$

$$\frac{dy}{dx} = \frac{1}{-\operatorname{cosec} y \sqrt{\operatorname{cosec}^2 y - 1}}$$

Using (i)

$$\frac{d}{dx}(\operatorname{cosec}^{-1} x) = -\frac{1}{x \sqrt{x^2 - 1}}$$

$$\frac{dy}{dx} = -\frac{1}{x\sqrt{x^2-1}}$$